

Day-1 Linear Regression

Given Data:

Load(N) x	Extension(mm) y
5	1.2
10	2.5
15	3.7
20	4.9
25	6.1

Equations

$$y = mx + c$$

$$m = \frac{n \sum xy - (\sum x)(\sum y)}{n \sum x^2 - (\sum x)^2}$$

$$c = \frac{\sum y - m \sum x}{n}$$
$$= \frac{18.4 - 18.3}{5}$$

$$c = 0.02$$

x	y	x ²	xy
5	1.2	25	6
10	2.5	100	25
15	3.7	225	55.5
20	4.9	400	98
25	6.1	625	152.5

$$\sum x = 75 \quad \sum y = 18.4 \quad \sum x^2 = 1375 \quad \sum xy = 337$$

$$m = \frac{5(337) - (75 \times 18.4)}{5(1375) - (75)^2} = \frac{1685 - 1380}{6875 - 5625}$$
$$= 305/1250$$

$$m = 0.244$$

$$y = 0.244x + 0.02$$

Prediction for 55 N:

$$y = 0.244 \times 55 + 0.02$$
$$= 0.02 + 13.42$$

$$y = \underline{\underline{13.44 \text{ mm}}}$$

Extension for 55 N = 13.44 mm



